

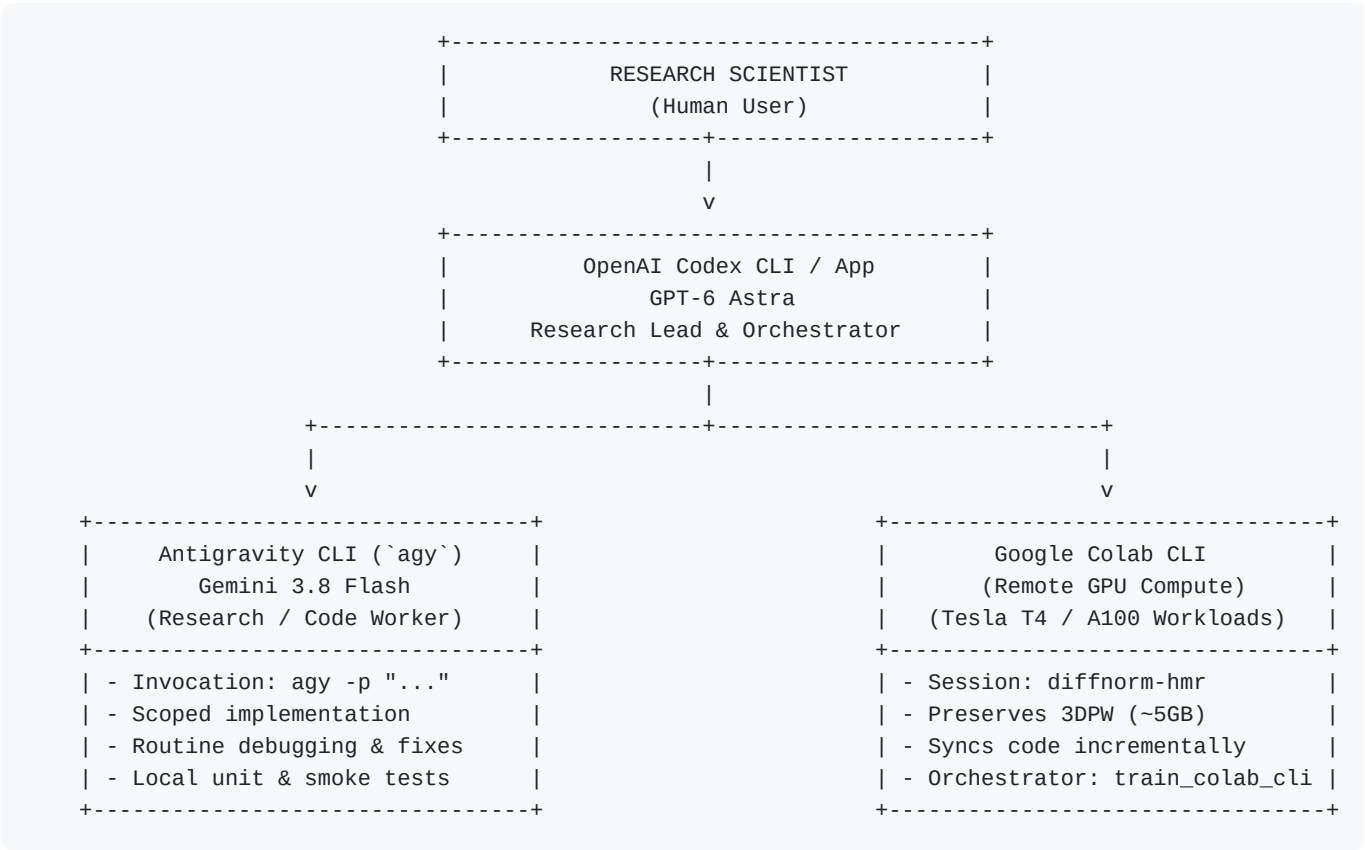
Subscription-Only Research Architecture Report: Astra + Gemini

Project: DiffNorm-Contact HMR
Date: September 17, 2026
Auditor / Engineer: Antigravity AI

1. Executive Audit & Architecture Transition

Area	Previous API/Coordinator Design	New Subscription-Only Codex Architecture	Verdict
Orchestration Runtime	Custom Python script (<code>research.py</code>) + <code>openai-agents</code> SDK	OpenAI Codex CLI / App (ChatGPT Plus)	Dramatically simplified — zero custom agent code to maintain or debug.
OpenAI Billing	Required <code>OPENAI_API_KEY</code> with per-token API billing	ChatGPT Plus subscription allowance	Zero API billing — uses personal plan quota in Codex.
Gemini Delegation	Python wrapper calling <code>subprocess.run(["agy", ...])</code>	Direct shell execution (<code>agy -p "..."</code>)	Zero overhead — uses signed-in Google account (Google AI Pro).
Colab Execution	Python tool wrapper	Direct CLI / Orchestration script (<code>train_colab_cli.sh</code>)	Native & idempotent — reuses <code>diffnorm-hmr</code> session and cached data.
Research Memory	Dispersed between script and files	Pure Git + Markdown (<code>RESEARCH.md</code> , <code>notes/</code>)	Auditable, persistent, and clean.

2. System Architecture



2.1 Role Division & Authority

- **GPT-6 Astra (Research Lead)** runs natively inside Codex.
 - Owns research questions, literature critique, falsifiable hypotheses, experiment design, controls/ablations, skeptical code review, and final scientific decisions.
 - Inspects every code change made by Gemini using `git status` and `git diff`.
- **Gemini 3.8 Flash (Research Worker)** is invoked directly from bash by Astra via `agy`.
 - Handles repository exploration, modular feature implementations, bug fixes, routine unit tests, and script preparation.
 - Claims made by Gemini are treated as unverified worker notes until audited.
- **Google Colab CLI (colab)** handles expensive remote GPU compute.

3. Directory Structure & Eliminated Artifacts

3.1 Files Completely Removed

The following files were eliminated to remove all API key dependencies and custom Python runtime overhead: - **✗** `research.py` (at root and in starter pack) - **✗** `run_research.sh` (at root and in `code/scripts/`) - **✗** `.env` and `.env.example` - **✗** `requirements.txt` / `openai-agents` dependency

3.2 Final Clean Structure

```
/home/dat/HMR/
├── AGENTS.md                # Unified system contract & project rules
├── RESEARCH.md              # Live high-level state (baseline, hypotheses, uncertainty)
├── .gitignore
├── notes/                   # Durable scientific memory
│   ├── literature.md        # Synthesized literature notes (DSINE, HMR2.0, etc.)
│   ├── hypotheses.md        # Falsifiable hypotheses (H001-H003)
│   └── decisions.md         # Decisions log (DEC-001 - DEC-004)
├── code/                   # Research implementation codebase
│   ├── src/                 # DiffNorm, Gaussian collision, and HMR models
│   ├── scripts/             # Colab orchestration & evaluation scripts
│   └── tests/               # Unit and smoke tests
├── documents/              # Documentation, review reports, and PDFs
│   ├── md/reviews/          # Markdown reports
│   └── pdf/                 # Publication-quality rendered PDFs
├── results/                # Machine-readable JSON metrics (pilot experiments)
└── (astra_gemini_research_starter removed: fully integrated into root)
```

4. Operational Protocols

4.1 Delegating Tasks to Gemini via `agy`

When Astra needs code exploration or implementation, Astra runs:

```
agy --model gemini-3.8-flash-high -p '
Task: Implement Condition-F stage-gated normal loss router.
Scope: Inspect code/src/optimization/capacity_scheduler.py and update gating threshold.
Constraints: Keep baselines untouched; do not run full training; add a unit test.
Report: Files changed, tests executed, and concise summary.
'
```

4.2 Reviewing Worker Changes

Immediately after Gemini finishes, Astra executes:

```
git status
git diff
pytest code/tests/ -q
```

Astra verifies that mathematical conventions, Lie group rotation formulations, and loss derivatives remain rigorous.

4.3 Launching Colab Experiments

Per repository rules in `AGENTS.md`, Astra executes the dedicated orchestration script:

```
bash code/scripts/train_colab_cli.sh
```

This guarantees: 1. Reusing the active `diffnorm-hmr` session without re-provisioning. 2. Preserving the uploaded 3DPW dataset (~5GB) and dependencies on the VM. 3. Synchronizing code incrementally. 4. Logging all execution steps to `colab_training.log`.

4.4 Ingesting Papers & Presenting Review Deliverables

- **Reading PDFs:** PyMuPDF4LLM is the default:

```
pymupdf4llm papers/example.pdf --stdout
```

- **Review Deliverables:** Mandatory PDF presentation using `render-pdf` :

```
render-pdf documents/md/reviews/report.md documents/pdf/report.pdf
```

5. Summary & Verification

- ✓ Removed all Python coordinator scripts (`research.py` , `run_research.sh` , `.env`).
- ✓ Zero API key dependencies (uses ChatGPT Plus in Codex and Google AI Pro in `agy`).
- ✓ Consolidated all starter assets into the repository root and eliminated redundant `astra_gemini_research_starter/` .
- ✓ Initialized `RESEARCH.md` and `notes/` with live DiffNorm-Contact HMR research state.